

CE 310 — Week 3 In-Class Exercise: Condition or Miss the Point

Dataset: DOE Reference Building — Medium Office, ASHRAE Climate Zone 2B (Tucson Area), 2023
Topic: Marginal Probability · Conditional Probability · Joint Probability · Independence

Before You Begin

Files needed:

- Week3_Tucson_Office_Energy_2023.csv — hourly building energy data (8,760 rows)
- Wk03.04_In_Class_Answer_Sheet.xlsx — download from D2L and enter all answers here

Dataset note: This is real simulation output from the U.S. Department of Energy's Reference Building program — a medium-sized office building (3 floors, ~53,600 sq ft) calibrated to Tucson's climate. Each row is one hour of 2023.

Answer Sheet instructions: Type every numerical answer in Col E of the row matching the question label — A1 in row A1, B3 in row B3. For ★ questions, also enter the intermediate value in Col D. For Col C, type the Excel formula you used.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761)'. Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted;** the typed formula is what earns the approach credit.

Background

This week you'll move from $P(\text{event})$ to $P(\text{event} \mid \text{condition})$ — and see how much a condition can change the picture.

Excel Quick Reference

Goal	Formula / Method
Count non-empty cells in a column	=COUNTA(A:A)-1 — subtract 1 to exclude the header row
Count hours meeting one condition	=COUNTIF(range, ">50")
Count hours meeting two conditions	=COUNTIFS(range1, ">50", range2, "Weekday")
Average for one group	=AVERAGEIF(B2:B8761, 7, G2:G8761) — average Elec where Month=7
Average for two conditions	=AVERAGEIFS(H2:H8761, E2:E8761, "Summer", F2:F8761, "Yes")
Binary indicator (0/1)	=IF(H2>50, 1, 0) — fills down for pivot table use
Insert pivot table	Insert tab → PivotTable → configure Rows, Values
Average in pivot table	Values area → Value Field Settings → Average

Goal	Formula / Method
Conditional probability	$= \text{joint count} / \text{group total} \times 100$
Independence test	$P(A \cap B) \text{ vs } P(A) \times P(B)$ — compute both and compare

Setup (~10 min)

You do all of this week's work in **one file: your Answer Sheet**. The data, your scratch work, and your charts all live there, so everything you built is in the file you hand in. Add as many sheets as you like — only Instructions, Answers, Work, Charts and Written have to keep their names.

1. Import **Week3_Tucson_Office_Energy_2023.csv** onto a sheet named Data that you add to your Answer Sheet:
 - Data → Get Data → From Text/CSV → Load.
2. Confirm **8,760 rows** of data (use $=\text{COUNTA}(A:A)-1$ to verify).
3. Confirm the **12 column names** in row 1:
 - DateTime, Month, Hour, DayType, Season, BusinessHours
 - Electricity_kWh, Cooling_kWh, HeatingElec_kWh, Lighting_kWh, Equipment_kWh, Gas_kWh
 - **DayType** = "Weekday" or "Weekend" · **Season** = "Winter/Spring/Summer/Fall"
 - **BusinessHours** = "Yes" (Mon–Fri 8am–6pm) or "No"
4. Add **two helper columns** — you will use these for the pivot table in Section B:
 - In **Column M**, header IsCoolHigh:
 $=\text{IF}(H2>50, 1, 0)$
 Fill down to row 8761.
 - In **Column N**, header IsElecHigh:
 $=\text{IF}(G2>100, 1, 0)$
 Fill down to row 8761.

Why 0/1? A binary indicator column turns a condition into a number. The average of a 0/1 column equals the fraction of rows where the condition is true — which is a probability. When you pivot on it, Excel computes conditional probabilities automatically.

Section A — Marginal Probabilities (~25 min)

Section A establishes the baseline — how often each event occurs across the whole year, with nothing held fixed. These are the numbers Section B will overturn.

Marginal probability = probability of an event without conditioning on anything else.

All Section A questions use COUNTIF or COUNTIFS on a single column. Round percentages and kWh values to 2 decimal places.

A1 (2 pts) — Hours With Any Cooling Load

How many hours in 2023 did the building use **any** cooling energy (Cooling_kWh > 0)?

This tells you how many hours the cooling system ran at all — including partial loads too small to matter for design.

A2 (2 pts) — Hours With High Cooling Demand

How many hours had Cooling_kWh > 50 — the threshold where the system is operating at a substantial fraction of design capacity?

A3 (2 pts) — Marginal P(Cooling > 50)

Compute the **marginal probability** that any randomly chosen hour has high cooling demand:

$$P(\text{Cooling} > 50) = \text{A2 value} / 8760 \times 100$$

Express as a percentage, rounded to 2 decimal places.

Section B shows this understates the risk at peak and overstates it in winter.

A4 (2 pts) — Hours With High Total Electricity

How many hours had Electricity_kWh > 100 — above the building's nominal base load?

A5 (2 pts) — Marginal P(Electricity > 100)

$$P(\text{Electricity} > 100) = \text{A4 value} / 8760 \times 100$$

A6 (2 pts) — Count of Summer Hours

How many of the 8,760 hours fall in Summer (Season = "Summer")?

A7 (2 pts) — Count of Weekday Hours

How many hours are on weekdays (DayType = "Weekday")?

A8 (2 pts) — July Average Electricity

Calculate the average hourly electricity demand for **July only** (Month = 7):

Compare this to the full-year mean (~87.4 kWh). July's higher average hints at what conditioning reveals in Section B.

Section B — Conditional Probabilities

Section B adds the condition. The same events, restricted to a month, a day type, or a schedule — and the gap between these answers and Section A's is the whole point of the week.

Report percentages to 2 decimal places, kWh values to 2 decimal places.

B1 (4 pts) — Joint Count: High Cooling AND July

How many hours had **both** Cooling_kWh > 50 **and** Month = 7 (July)?

B2 ★ (5 pts) — P(Cooling > 50 | July)

The **conditional probability** of high cooling demand given that the month is July:

$$P(\text{Cooling} > 50 \mid \text{July}) = (\text{hours with Cooling} > 50 \text{ AND Month} = 7) / (\text{total July hours}) \times 100$$

Col D: total July hours (=COUNTIF(B2:B8761, 7)) · **Col E:** P(Cooling > 50 | July) as %

Divide by your A3 marginal: that ratio is how much knowing the month changes the answer.

B3 (4 pts) — P(Cooling > 50 | January)

Repeat for January (Month = 1):

A zero here is real, not a broken formula. January electricity still averages ~82 kWh/hr — lighting and plug loads, not cooling.

Complement rule: $P(\text{Cooling} \leq 50 \mid \text{January}) = 1 - B3$. No computation needed — state the value.

B4 (4 pts) — Independence Test: Cooling Demand and Month

If high cooling demand were **statistically independent** of month, then:

$$P(\text{Cooling} > 50 \text{ AND July}) \text{ would equal } P(\text{Cooling} > 50) \times P(\text{July})$$

Compute this **independence product**:

$$= (\text{A3 value} / 100) \times (\text{July_hours} / 8760) \times 100$$

where July_hours = **744**, your B2 intermediate — not A6, which counts Summer's 2,208. Express as a percentage.

If this independence product matches the actual joint probability $P(C > 50 \text{ AND July}) = B1/8760 \times 100$, the variables are independent. If not, conditioning on month changes the probability — and the variables are **associated**.

B5 (4 pts) — Joint Count: High Electricity AND Weekday

How many hours had **both** Electricity_kWh > 100 and DayType = "Weekday"?

B6 ★ (5 pts) — P(Electricity > 100 | Weekday)

$$P(E > 100 \mid \text{Weekday}) = \text{B5 value} / \text{total weekday hours} \times 100$$

Col D: total weekday hours (your A7) · **Col E:** P(E > 100 | Weekday) as %

Engineering Context: NFPA 110 requires backup generators sized for the maximum anticipated connected load — not the average load.

B7 (4 pts) — P(Electricity > 100 | Weekend)

Compare B6 to B7. The ratio $P(E>100|Weekday) / P(E>100|Weekend)$ is the **lift** — a metric used in data science to measure how much a condition changes a probability. A lift > 1 means positive association; lift < 1 means negative association.

B8 (4 pts) — Independence Product: Electricity and Day Type

Compute the **independence product** — what $P(\text{Electricity} > 100 \text{ AND } \text{Weekday})$ would be if the two events were unrelated:

$$\text{Independence product} = (\text{A5 value} / 100) \times (\text{A7 value} / 8760) \times 100$$

For comparison only — do not enter it — the actual joint probability is $B5 / 8760 \times 100$.

If the independence product equals the actual joint probability, the events are independent. If they differ, they are associated. Which is larger here, and what does that tell you about whether high electricity demand is more or less common on weekdays than chance would predict?

Addition rule check (ungraded): Compute $P(\text{Electricity} > 100 \text{ OR } \text{Weekday}) = P(E>100) + P(\text{Weekday}) - P(E>100 \text{ AND } \text{Weekday})$, where $P(\text{Weekday}) = A7/8760 \times 100$ and $P(E>100 \text{ AND } \text{Weekday}) = B5/8760 \times 100$. Show each term in the Work sheet's Addition-rule check block, B32 to B35.

B9 ★ (5 pts) — P(Electricity > 100 | Business Hours)

How likely is high electricity demand during official business hours (Mon–Fri 8am–6pm)?

Col D: business-hours count ($=\text{COUNTIF}(F2:F8761, "Yes")$) · **Col E:** $P(E > 100 | \text{BusinessHours} = \text{Yes})$ as %

B10 (4 pts) — P(Electricity > 100 | NOT Business Hours)

B9 against B10 is the result this in-class exercise is built on: sizing to the marginal 39.3% (A5) over-provisions off-hours and under-provisions business hours at the same time.

ArcE Framing: ASHRAE 90.1 requires HVAC systems sized for design conditions, not average conditions. A system sized for the annual average cooling demand (18.96 kWh/hr) would be severely undersized during summer business hours.

B11 (4 pts) — Pivot Table: P(Cooling > 50 | Season)

Use the IsCoolHigh helper column (Col M) to build a pivot table:

Step 1 — Insert Pivot Table: 1. Click any cell in your data. **Insert tab** → **PivotTable** → choose “New Worksheet” → OK.

Step 2 — Configure fields: - Drag **Season** to the **Rows** area. - Drag **IsCoolHigh** to the **Values** area. - In the Values area, click the dropdown → **Value Field Settings** → change from “Sum” to “Average”.

Step 3 — Read the result: The pivot table now shows the average of IsCoolHigh for each season. Since IsCoolHigh is 0 or 1, the average equals the fraction of hours with Cooling > 50 — which is $P(\text{Cooling} > 50 | \text{Season})$.

Multiply each value by 100 to convert to percentage. Enter the **Summer** value as your answer.

Alternative (no Pivot Table required): If your version of Excel does not support Pivot Tables, use COUNTIFS to divide the count of Summer hours with Cooling > 50 kWh by the total count of Summer hours. The result is identical to the pivot table average.

B12 (4 pts) — Month With Highest Conditional Cooling Probability

Extend the same pivot table: drag **Month** into the Rows area in place of **Season**, and it reports all twelve months. - **Copy those twelve numbers into Work!B7:B18**, beside the month labels already there. Chart 1 plots that block, so it needs the values whether you pivoted or not — a pivot table Excel drops on an auto-named sheet is not somewhere a preceptor can find them.

Which month (enter the month number) has the **highest** conditional probability of high cooling demand?

Alternative (no Pivot Table required): The Work sheet of your Answer Sheet has the twelve months already listed in the block headed *Chart 1 + question B12*. Compute $P(\text{Cooling} > 50 \mid \text{Month} = m)$ into B7 down to B18 using COUNTIFS — the count of hours in that month with Cooling > 50 kWh, divided by the total count of hours in that month, times 100. The row with the largest value gives your answer.

B13 (4 pts) — Mean Cooling During Peak Conditions

Calculate the average cooling demand during the highest-load hours: **Summer** season **and** business hours:

Against the annual mean of ~19 kWh/hr, the ratio is what sizing to $P(\text{event})$ instead of $P(\text{event} \mid \text{design condition})$ costs you.

B14 ★ (5 pts) — Probability Bridge: $P(\text{Weekday} \mid \text{July})$

So far you have computed $P(\text{event} \mid \text{time condition})$. Now reverse the conditioning:

$P(\text{Weekday} \mid \text{July}) = ?$

Col D: count of Weekday AND July hours · **Col E:** ÷ total July hours (from B2) × 100

Notice: $P(\text{Weekday} \mid \text{July}) \neq P(\text{Weekday})$ overall. July has the same weekday/weekend structure as every other month — so why do they differ? They shouldn't differ by much if the calendar is symmetric. In this week's lecture you learned **Bayes' theorem** for reversing conditional probabilities systematically: $P(B|A) = P(A|B) \cdot P(B) / P(A)$. You can verify your B14 answer two ways — directly by COUNTIFS and by applying Bayes to your B2 result — and both should match. This is the foundation of probabilistic reasoning and of the Naive Bayes classifier.

Where the working goes. Your Answer Sheet has a **Work** sheet with a labelled, pre-filled block for each table below. Build them there and nowhere else — the charts are scored by looking at the cells they plot, so a table in a blank corner of another sheet is a table your preceptor has to hunt for.

Required Charts (15 pts)

Both charts belong on the **Charts sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build the chart wherever your data is, then right-click it -> **Move Chart** -> **Object in** -> Charts, and drag it into the matching box. Leave it as a live chart — do not replace it with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (7 pts) — Monthly Conditional Probability Profile

From your pivot table (Month as rows, Average of IsCoolHigh as values), build a bar chart. Multiply y-values by 100 to show percentages. Add a horizontal reference line at your A3 marginal value.

- Title: $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Month})$ – Tucson Office Building 2023
- Axes: Month (1–12) on x, Conditional Probability (%) on y

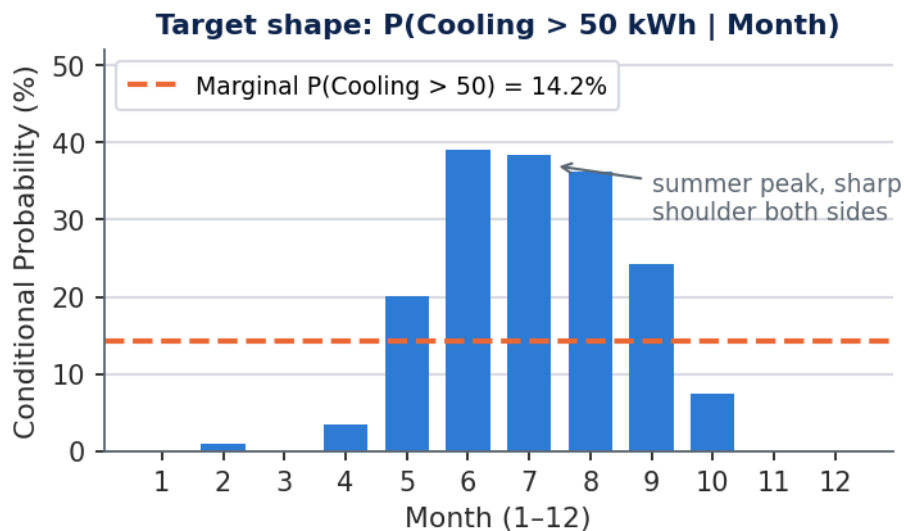


Figure 1: Target shape for Chart 1

Chart 2 (8 pts) — Conditioning Effect Comparison

The Work sheet has the five conditions listed in order in the block headed *Chart 2* — Marginal (all hours), Weekday, Weekend, Business Hours, Non-Business Hours. Copy each probability from your Section B answers into B23 down to B27, then plot that block as a clustered bar chart with data labels on each bar. Keeping the given order earns the *bars ordered logically* point.

- Title: How Conditioning Changes $P(\text{Electricity} > 100 \text{ kWh})$ – Tucson Office 2023
- Axes: condition labels on x, Probability (%) on y

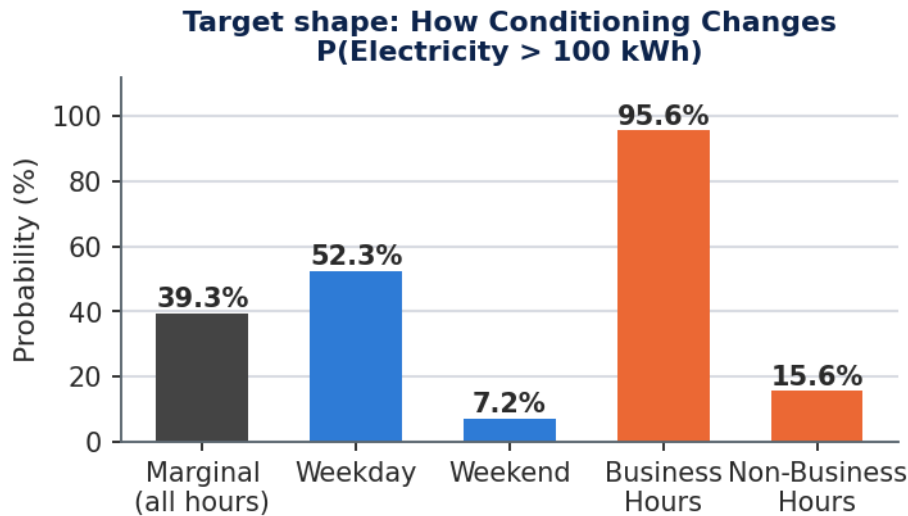


Figure 2: Target shape for Chart 2

Memo Deliverable (9 pts)

Type your memo directly in the **Written sheet** of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed.

To: CE 310 Instructor
 From: [Your Name], [Your Major]
 Date: [Date]

Subject: Week 3 In-Class Exercise – Conditional Probability Analysis, Tucson Office Building 202

Paragraph 1 (3 pts) — Marginal vs. Conditional: State the marginal $P(\text{Cooling} > 50 \text{ kWh})$ and $P(\text{Electricity} > 100 \text{ kWh})$. Then describe how conditioning on month or time of day changes these probabilities. Cite at least two specific conditional probability values from your Section B results. Explain in one sentence why the marginal probability is misleading for design purposes.

Paragraph 2 (3 pts) — The Conditioning Effect: Using your Chart 2, describe the range of $P(\text{Electricity} > 100 \text{ kWh})$ across your five conditions. What does the ratio $P(E > 100 | \text{BusinessHours}) / P(E > 100 | \text{NonBusinessHours})$ tell you about the relationship between occupancy schedule and electricity demand? Reference the independence test result from B4 or B8.

Paragraph 3 (3 pts) — Cooling system sizing and conditional probability:

An HVAC engineer is sizing a chilled-water cooling system for a medium office building in Tucson. Using your B11, B12, and B13 results, explain why the ASHRAE 90.1 design conditions require conditioning on season and time of day, not just annual average cooling load. Cite at least one specific conditional probability from your pivot table.

Submission: Save as CE310_W3_<NetID>.xlsx and upload to the Week 3 D2L dropbox. Enter your NetID in the **NetID** cell at the top of the Answers sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
 - ☐ File saved as .xlsx and named CE310_W3_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in **Col D**
 - ☐ Both charts sitting in their boxes on the Charts sheet, as live charts and not pictures, with titles and axis labels
 - ☐ Memo written in the Written sheet
 - ☐ Uploaded to D2L by **9:00 AM next Wednesday**
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Grading Summary

Component	Points
Section A — A1–A8	16
Section B — B1–B14	60
Chart 1 — Monthly conditional probability profile	7
Chart 2 — Conditioning effect comparison	8
Memo (3 paragraphs)	9
Total	100